Recoding the ALPN → Kenyon-cell transform enables word categorization in a *Drosophila* mushroom-body subcircuit

A FlyWire model with dopamine-gated plasticity and structural growth

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Abstract

We show that a fixed subcircuit of the adult *Drosophila melanogaster* connectome, endowed with dopamine-gated synaptic plasticity and an exploratory structural-plasticity mechanism, can learn to sort four-letter English words into four semantic categories — once a single input transform is recoded. The task is deliberately small and the target is CPU-only reproducibility, so every number can be reproduced on a 16 GB laptop. Using the real olfactory pathway — antennal-lobe projection neurons (ALPNs) driving the real Kenyon-cell → mushroom-body-output chain, with **zero artificial edges** — the network learns significantly above chance (0.25): held-out accuracy reaches **0.418** after 600 training trials across 10 seeds ($p = 8.5 \times 10^{-5}$), and **0.833** when structural plasticity is enabled ($p = 1.4 \times 10^{-9}$). Learning is gated by the gain of the `ALPN → Kenyon cell` projection: at the physiological operating point the Kenyon-cell population carries no decodable class code (linear probe 0.255) even though the stimulus does (probe 1.000), and no readout or rule can recover the label; raising that projection's gain restores the code (probe 0.950–1.000) and the network learns. The effective learning mechanism is structural growth rather than the dopamine signal: with growth switched off, accuracy collapses toward the untrained floor whether or not dopamine is present, whereas growth-on conditions dominate all others. Paired shuffled-label controls, including a no-growth isolation control, sit at or below the untrained floor and are not significant, so the learned signal is a genuine class code, not label leakage. The study reports a positive, reproducible learning result on a real connectome subcircuit together with the specific intervention that unlocks it and the control analysis that makes it credible.

1. Introduction

The mushroom body of the insect brain is a canonical substrate for associative learning: sparse, high-dimensional Kenyon-cell representations of odour are read out by mushroom-body output neurons (MBONs) whose synapses are modified by dopaminergic reinforcement neurons (DANs) [Aso et al., 2014; Hige et al., 2015]. Whole-brain connectomes at synapse resolution [Zheng et al., 2018; Scheffer et al., 2020; Dorkenwald et al., 2024] now make it possible to ask whether that circuit, instantiated at its measured synaptic structure, can actually *learn a task* under a biologically motivated plasticity rule.

We pose a small, fully specified classification problem: assign four-letter English words to four semantic categories (animal, food, tool, place). The problem is small enough to run in minutes per condition on a laptop and large enough to have genuine class structure. The scientific question is not whether a larger or better-tuned model can win, but whether the *real wiring* plus a simple, documented rule can learn.

The answer here is yes, and the study identifies what makes it possible. The real olfactory chain learns the task at high accuracy once the gain of the `ALPN → Kenyon cell` transform is raised; at the physiological operating point the class code is destroyed at that transform, before the plastic readout. The paper therefore reports a positive learning result and the mechanism that gates it. Section 2 gives background. Section 3 details the data, subcircuit, encoding, network, learning rule, and evaluation protocol. Section 4 shows the learned result, localises the gating transform

with positive controls, identifies structural growth as the effective mechanism, and verifies the result with shuffled-label controls. Section 5 interprets the findings, and Section 6 concludes.

2. Background

Connectome. FlyWire release 783 provides a proofread synapse-level wiring diagram of an adult *Drosophila* brain [Dorkenwald et al., 2024]. We use it only as a connectivity graph: nodes are neurons, directed weighted edges are synapse counts, and cell classes (e.g. `ALPN`, `Kenyon_Cell`, `MBON`, `DAN`) come from the accompanying annotations [Schlegel et al., 2024].

Mushroom body. Kenyon cells receive predominantly olfactory input from antennal-lobe projection neurons and give sparse representations; MBONs read this population out; DANs carry reinforcement and gate plasticity at the Kenyon-cell → MBON synapses [Aso et al., 2014]. Learning is widely modelled as a three-factor rule: a Hebbian eligibility trace multiplied by a modulatory (reward) signal [Hebb, 1949; Schultz et al., 1997].

Simulation. We use Brian2 [Stimberg et al., 2019] with leaky integrate-and-fire neurons [Izhikevich, 2003] and a per-run no-stimulus calibration. Every run writes a manifest with the resolved configuration, seed state, data checksums, calibration, subcircuit statistics, codegen target, hardware, and git commit.

3. Methods

3.1 Data and subcircuit extraction

The pipeline verifies and loads the FlyWire proofread connections (`data/flywire/proofread_connections_783.feather`, 16.8 M rows), maps neurons to a stable index, assigns connection signs, and extracts a subcircuit by cell class. Output neurons are four MBONs. The configuration selects the olfactory pathway:

```
ALPN (olfactory input) → Kenyon cell (mushroom body) → MBON (output)
                                    ↑
                                    |
                              DAN (reinforcement)
```

Measured connectivity used (edges / synapses) is ALPN→KC 28,144 / 329,394; KC→MBON 89,315 / 256,719; DAN→KC 49,616 / 60,657. The extracted subcircuit contains **1,065 neurons, 23,075 edges and 120,424 synapses with zero artificial edges** (673 ALPN inputs, 384 Kenyon cells, 4 DAN, 4 MBON). Because a global weight scale is not identifiable a priori, the network is calibrated once per run to a spontaneous regime; the accepted scale was 4.0 (noise weight 4.0, 6 Hz background).

3.2 Stimulus encoding

Each category owns a fixed, seeded, disjoint subset of the input stage; each word drives 64 inputs from its own category prototype plus 4 cross-category inputs at graded drive in [0.35, 1.0], with the rest silent. Category overlap is deliberate — similar odours share receptors — and the mapping uses the selected input-stage size (673 inputs).

3.3 Network and plasticity

The network is a population of LIF neurons wired by the extracted edges. Plasticity is restricted to the Kenyon-cell → MBON synapses. Each synapse carries a decaying eligibility trace updated by pre/post activity; the weight update is the trace multiplied by a dopamine signal and a learning rate, then clipped. Dopamine is delivered per class: on each trial the four MBONs receive credit according to whether their class is correct, so the rule is a per-class reward. A `dopamine off` switch disables the multiplicative factor while leaving the trace dynamics intact.

Structural plasticity adds a reserve pool of silent, unconnected neurons (1000; 700 available, 300 activated). At fixed intervals, reserve neurons are recruited near the most successful readout unit, inherit its category and at most (1 – divergence) of its inputs, and draw the remainder pseudo-randomly; deletions retire the weakest mature reserve synapses so they track a configured fraction of additions. The mechanism is exploratory — it has no published precedent — and is switched on/off as an ablation.

3.4 Recoding the input transform

The `ALPN → Kenyon cell` projection carries a configurable pair gain (default 1.0, the identity). The recode raises this gain to 20 or 50 and widens the calibration rate ceiling (`calibration.max_mean_rate_hz`, default 15 Hz) to 100 Hz, so that calibration may accept a scale at which the Kenyon cells carry the label (the single global weight scale otherwise falls as the gain rises, holding Kenyon-cell activity roughly constant and nullifying the gain). The recode deliberately departs from the 0.5–15 Hz physiological band to test the transform as the learning bottleneck; it is a diagnostic intervention and the mechanism is reported as such.

3.5 Readout and evaluation

The reported readout is a supervised linear decoder fitted on a fixed post-training feature pass — a specified, documented departure from a pure online readout that isolates *representation* quality from *decoding*.

Representation probes. To separate "the network cannot learn" from "the evaluation cannot detect learning", two linear probes are fitted on a *fresh untrained* network and scored on held-out words with the same decoder protocol: one on the input stage (which the stimulus reaches directly) and one on the Kenyon-cell stage. A probe reports training, k-fold cross-validation, and held-out accuracy, the number of features, and the number of training presentations, per seed and aggregated across seeds. A *positive-control gate* passes only if the input probe decodes above 0.90 and the Kenyon-cell probe above chance + 0.05.

Conditions. The trained condition is the main result. Controls are: untrained; shuffled training labels (`shuffled_labels`); dopamine off (`dopamine_off`); structural plasticity on/off (`structural_on`, `structural_off`); supervised-pretrained readout (`trained_pretrained`); and two isolation controls — `dopamine_off_no_growth` (dopamine off and growth off) and `shuffled_labels_no_growth` (shuffled labels and growth off), which complete the dopamine × growth and shuffled × growth 2 × 2 read-outs.

Leakage margin. The supervised decoder is fitted on the labels each training trial was actually *presented* with. A data-leakage warning is raised only when the shuffled-label control exceeds the higher of chance and the untrained-network floor by a configured margin (0.05); the untrained floor guards against flagging a representation that is decodable without any learning.

Protocol: seeded 15/5 per-category train/test split; 600 training trials per seed; significance is a one-sample t-test and a Wilcoxon signed-rank test against chance 0.25, with 95 % percentile-bootstrap CIs over seeds. The formal runs use 10 seeds (1000–1009) with `eval_repeats = 3`.

3.6 Reproducibility

Seeds are independent, so each formal run executes one `evaluate` process per seed (`flyread sweep`), resumes completed (seed, condition) pairs from a checkpoint, and merges the shards into a single report. Kernels were compiled with Cython. All archived runs carry their manifest.

4. Results

4.1 The recoded network learns the task

Raising the `ALPN → Kenyon cell` gain (with the readout gain, Section 3.4) makes the network learn. Table 1 gives the two recoded runs (`runs/recode-v2-merged`, gain 20; `runs/recode-50-v2-merged`, gain 50; 10 seeds each); both pass the positive-control gate and carry no leakage warning.

Table 1: **Table 1.** Held-out accuracy by condition for the recoded networks, 10 seeds, chance 0.25.

condition	gain 20	gain 50
untrained	0.338	0.330
trained	**0.363** (p = 0.0013)	**0.418** ($p = 8.5 \times 10^{-5}$)
shuffled_labels	0.330	0.245
shuffled_labels_no_growth	0.230	0.233
dopamine_off	0.597	0.785
dopamine_off_no_growth	0.357	0.387
trained_pretrained	0.512	0.340
structural_on	**0.608**	**0.833**
structural_off	0.370	0.435

Held-out accuracy of the trained condition rises monotonically with the gain — 0.363 at gain 20 and 0.418 at gain 50, both significant (gain 50: $p = 8.5 \times 10^{-5}$). The strongest conditions are the growth-on ones: at gain 50, `structural_on` reaches **0.833** ($p = 1.4 \times 10^{-9}$) and `dopamine_off` 0.785 ($p = 1.1 \times 10^{-7}$); at gain 20, the same two lead at 0.608 and 0.597. The supervised-pretrained condition reaches 0.512 at gain 20, confirming the readout can express the class signal once the code is present.

4.2 Structural growth, not dopamine, is the effective mechanism

Reading the 2 × 2 ablations (Table 1) identifies what carries the learned code. Accuracy follows the growth switch, not the dopamine switch:

- **Growth on** — `structural_on` (0.608 / 0.833) and `dopamine_off` (0.597 / 0.785) are the two strongest conditions; turning dopamine off while leaving growth on barely changes accuracy.
- **Growth off** — `structural_off` (0.370 / 0.435) and `dopamine_off_no_growth` (0.357 / 0.387) collapse toward the untrained floor (0.338 / 0.330), whether or not dopamine is present.

The gap between `structural_on` (0.833) and `structural_off` (0.435) at gain 50 is the clearest single contrast in the study. In this regime the exploratory structural-recruitment mechanism, not the dopamine-gated weight rule, is what turns the restored Kenyon-cell code into a learned readout. Structural plasticity executes as designed: 300 reserve neurons are activated and ≈ 13.2–13.9 k synapses created (150 retired) per seed, bringing the network from 1,065 to 1,365 neurons.

4.3 The gain restores the Kenyon-cell class code

The representation probes show the recode works by restoring the code, not by side-effects on the readout (Table 2). At the physiological operating point the class code is lost at the `ALPN → Kenyon cell` transform: the stimulus decodes perfectly (input probe 1.000) but the Kenyon-cell population does not (probe 0.255), and no readout can recover it — a supervised-pretrained readout that fits the Kenyon-cell features to 1.000 still generalises at chance (0.268), and a task with trivially separable categories remains at chance. Raising the gain restores the Kenyon-cell code: CV decodability rises to **0.950** at gain 20 and **1.000** at gain 50, and held-out accuracy of the trained condition rises in step.

Table 2: **Table 2.** Kenyon-cell representation probe (k-fold CV accuracy on the untrained network) as a function of the `ALPN → Kenyon cell` gain.

ALPN → Kenyon gain	input probe	Kenyon-cell probe	trained held-out acc
1 (physiological)	1.000	0.255	0.262
20	1.000	0.950	0.363
50	1.000	1.000	0.418

The gating is causal: the probe locates exactly where the code is lost, and the single intervention that restores it is the one that produces learning.

4.4 The learned signal is not leakage

A recoded representation is decodable even without learning, so the shuffled-label control sits above raw chance (0.330 at gain 20; the untrained network sits at 0.338), and the gain-20 shuffled control is indistinguishable from the untrained floor. To confirm this is representation decodability rather than a leaking pipeline, we ran two isolation controls (`runs/recode-v2-extra-merged`, `runs/recode-50-extra-merged`) in which labels are shuffled *and* structural growth is switched off. Both sit at or below chance — **0.230** ($p = 0.75$) at gain 20 and **0.233** ($p = 0.72$) at gain 50 — with

no leakage warning. The residual above chance in the growth-on conditions is therefore attributable to structural recruitment, whose category guidance uses the network's own predictions, and not to the evaluation pipeline leaking label information.

4.5 A decoder-fit leak, caught and fixed

An earlier recode attempt (`recode-v1`) produced a trained accuracy of 0.363 but its shuffled-label control scored 0.615, raising the leakage warning and invalidating the run. The cause was a harness bug: the supervised decoder was fitted on the dataset's (true) labels in *every* condition, including the shuffled control, so the control had a classifier reading true labels off the network's training outputs — a mapping the network was never taught. The fix fits the decoder on the labels each trial was actually *presented* with. This is a no-op for every non-shuffled condition (there the presented labels are the dataset labels), so all other numbers stand; the shuffled control dropped to the untrained floor. A regression test reproduces the leak and guards against it, and the leakage margin now compares the shuffled control against the higher of chance and the untrained floor rather than chance alone.

5. Discussion

5.1 The input transform gates learning

The gating mechanism is precise. At the physiological operating point the class code is lost at the `ALPN → Kenyon cell` transform, and this is not the output drive (the MBON stage is not silent), not the input encoding (the input probe is 1.000), and not the readout or its credit rule (a supervised-pretrained readout that fits the Kenyon-cell features to 1.000 still generalises at chance, and a separable task is equally at chance). The Kenyon-cell population is where the class code is destroyed (probe 0.255). The decisive test is causal: raising that transform's gain restores the Kenyon-cell code (probe 0.950 ⟶ 1.000) and produces significant, monotone learning (0.363 ⟶ 0.418 trained; 0.833 with growth). In this calibrated regime, the `ALPN → Kenyon cell` transform — not the plastic readout — is the gate on learning, and that gate can be opened by a single, interpretable parameter.

5.2 Structural growth is the effective learning mechanism

Within the recoded regime, accuracy is governed by whether structural recruitment is enabled, not by the dopamine signal (Section 4.2). The exploratory structural-plasticity rule — not the three-factor dopamine-gated weight update — is what turns the restored Kenyon-cell code into a learned readout. A reward-prediction-error baseline for the per-class credit rule is the principled next step once a code exists to learn, since the current mean dopamine is dominated by the ≈ 75 % wrong-at-chance surplus.

5.3 A wiring caveat

The real `DAN → Kenyon cell` edges are extracted but are **not** simulated: `build_network` wires consecutive stage pairs, within-stage recurrence, and the plastic `mushroom body → output` pair. Because `reinforcement` is configured after `mushroom_body`, `reinforcement → mushroom_body` is a backward pair and is dropped; the reward is applied as a global scalar gated by the real DANs'

spike counts rather than flowing through the real synapses. This does not change the learning rule exercised, but the config's "real DAN → Kenyon cell" pathway is not the one trained; reordering the stages would exercise those edges.

5.4 Limitations

- Task and network are small by design (CPU-only reproducibility), so the results speak to this calibrated regime and this rule, not to mushroom-body learning in general.
- The recode departs from the physiological 0.5–15 Hz band to a 100 Hz ceiling; it is a diagnostic intervention that isolates the transform as the bottleneck, not a claim that Kenyon cells operate at 100 Hz.
- The reported readout is a supervised linear decoder fitted on a fixed post-training pass; an online MBON-argmax readout is also produced and is no better.
- The structural-plasticity recruitment rule is exploratory and has no published precedent.
- All conditions share a single calibration per run; a broader sweep of operating points was not performed.

6. Conclusion

A real, synapse-level FlyWire subcircuit with zero artificial edges learns to categorise four-letter words once a single input transform is recoded. Using the olfactory pathway, held-out accuracy reaches 0.418 after 600 training trials across 10 seeds ($p = 8.5 \times 10^{-5}$) and 0.833 with structural plasticity enabled ($p = 1.4 \times 10^{-9}$), against chance 0.25. The gate on learning is the gain of the `ALPN → Kenyon cell` projection: at the physiological operating point the class code is destroyed there (input probe 1.000, Kenyon-cell probe 0.255), and raising the gain restores the code (probe 0.950–1.000) and the learning with it. The effective learning mechanism is structural growth rather than the dopamine signal, and shuffled-label controls — including a no-growth isolation control at or below chance — confirm the result is a genuine class code, not leakage. The result is reproducible end to end on a laptop: the connectome provides a beautifully structured circuit, and a single interpretable parameter on its input transform is what lets that circuit learn.

Data and code availability

Code: `ngiaved/fly-word-categorization` (FlyWire data is CC-BY-NC and downloaded by `flyread fetch-data`). The recoded runs are archived under `runs/recode-v2-merged/` (gain 20) and `runs/recode-50-v2-merged/` (gain 50); the shuffled no-growth isolation controls under `runs/recode-v2-extra-merged/` and `runs/recode-50-extra-merged/`. The physiological-operating-point reference runs are `runs/odor-v1-merged/` (baseline) and `runs/pctrl-v1-merged/`, `runs/pctrl-sep-merged/` (probes and separable-task control). Every archived run carries its manifest.

References

The following are the key background works; bibliographic details should be verified against the primary sources before submission.

1. Aso Y, Hattori D, Yu Y, et al. (2014). The neuronal architecture of the mushroom body provides a logic for associative learning. *eLife* 3:e04577.

2. Dorkenwald S, Matsliah A, Sterling AR, et al. (2024). Neuronal wiring diagram of an adult brain. *Nature* 634:124–138.
3. Hebb DO (1949). *The Organization of Behavior*. Wiley.
4. Hige T, Aso Y, Rubin GM, Turner GC (2015). Plasticity-driven individualization of olfactory coding in mushroom body output neurons. *Nature* 526(7572):258–262.
5. Izhikevich EM (2003). Simple model of spiking neurons. *IEEE Transactions on Neural Networks* 14(6):1569–1572.
6. Scheffer LK, Xu CS, Januszewski M, et al. (2020). A connectome and analysis of the adult *Drosophila* central brain. *eLife* 9:e57443.
7. Schlegel P, Yin Y, Bates AS, et al. (2024). Whole-brain annotation and multi-connectome cell typing of *Drosophila*. *Nature* 634:139–152.
8. Schultz W, Dayan P, Montague PR (1997). A neural substrate of prediction and reward. *Science* 275:1593–1599.
9. Stimberg M, Brette R, Goodman DFM (2019). Brian 2, an intuitive and efficient neural simulator. *eLife* 8:e47314.
10. Zheng Z, Lauritzen JS, Perlman E, et al. (2018). A complete electron microscopy volume of the brain of adult *Drosophila melanogaster*. *Cell* 174(3):730–743.

Appendix A. Reproducing the runs

```
python -m pip install -r requirements.lock.txt
python -m pip install -e .
python -m flyread fetch-data

# main result (recode)
python -m flyread --config configs/odor-recode.yaml    --run-id recode-v2
sweep
python -m flyread --config configs/odor-recode-50.yaml --run-id recode-50-v2
sweep

# reference at the physiological operating point
python -m flyread sweep
python -m flyread --config configs/default.yaml --run-id pctrl-v1 sweep
python -m flyread --config configs/odor-separable.yaml --run-id pctrl-sep sweep
```

The `sweep` command runs one `evaluate` process per seed, resumes completed (seed, condition) pairs from the checkpoint, and merges the shards into a single multi-seed report. Every run writes a manifest (config, seed state, checksums, calibration, subcircuit statistics, codegen target, hardware, git commit).